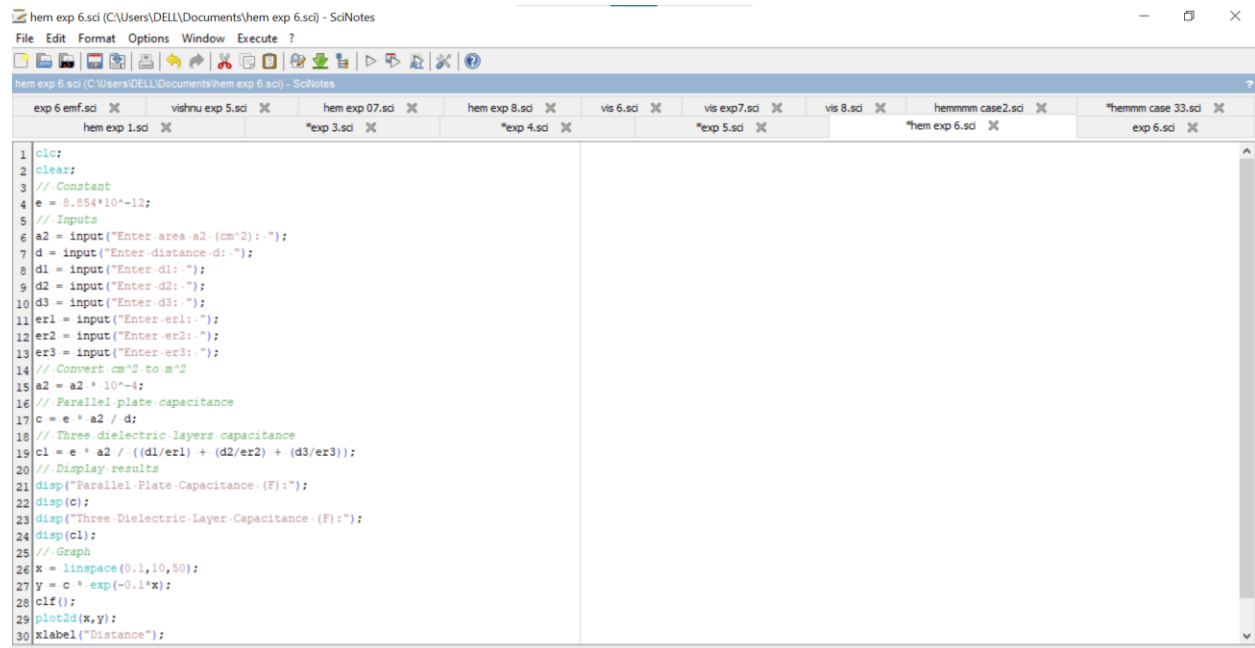
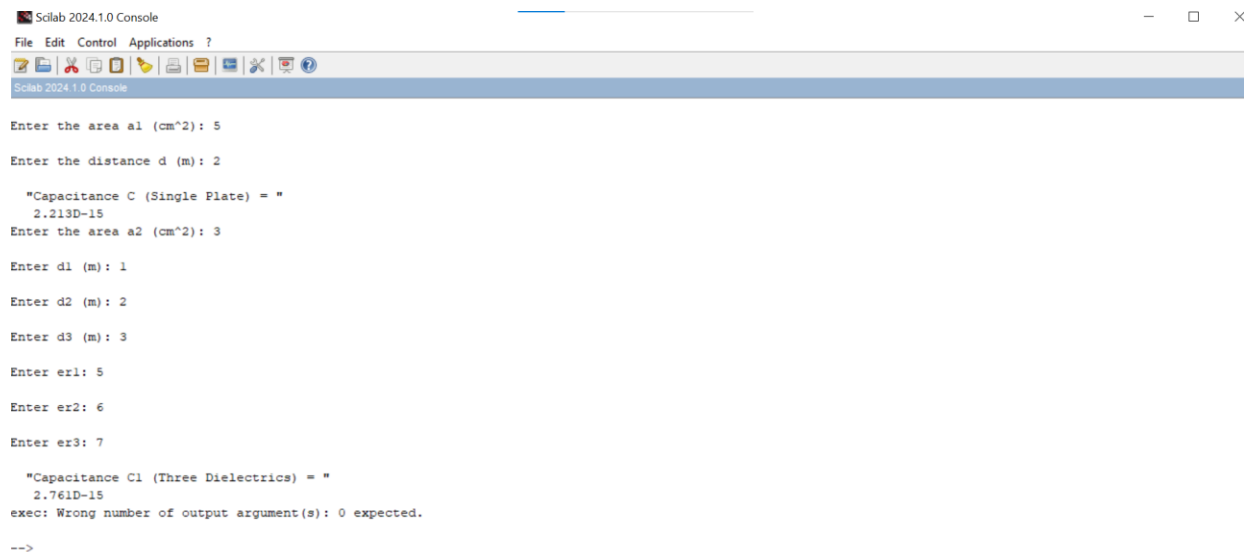


EXP 6: Determination and plotting of Capacitance of parallel plate capacitor in three different dielectric medium with specific area, dielectric thickness and various dielectric constant.



```
1 clc;
2 clear;
3 //Constant
4 e = 8.854*10^-12;
5 //Inputs
6 a2 = input("Enter area a2 (cm^2): ");
7 d = input("Enter distance d: ");
8 d1 = input("Enter d1: ");
9 d2 = input("Enter d2: ");
10 d3 = input("Enter d3: ");
11 er1 = input("Enter er1: ");
12 er2 = input("Enter er2: ");
13 er3 = input("Enter er3: ");
14 // Convert cm^2 to m^2
15 a2 = a2 * 10^-4;
16 // Parallel plate capacitance
17 c = e * a2 / d;
18 // Three dielectric layers capacitance
19 c1 = e * a2 / ((d1/er1) + (d2/er2) + (d3/er3));
20 // Display results
21 disp("Parallel Plate Capacitance (F):");
22 disp(c);
23 disp("Three Dielectric Layer Capacitance (F):");
24 disp(c1);
25 // Graph
26 x = linspace(0.1,10,50);
27 y = c * exp(-0.1*x);
28 clf();
29 plot2d(x,y);
30 xlabel("Distance");
```



```
Scilab 2024.1.0 Console

File Edit Control Applications ?

Enter the area a1 (cm^2): 5

Enter the distance d (m): 2

"Capacitance C (Single Plate) = "
2.213D-15
Enter the area a2 (cm^2): 3

Enter d1 (m): 1

Enter d2 (m): 2

Enter d3 (m): 3

Enter er1: 5

Enter er2: 6

Enter er3: 7

"Capacitance C1 (Three Dielectrics) = "
2.761D-15
exec: Wrong number of output argument(s): 0 expected.

-->
```

Exp 06:-

Enter the a_1 value

$$d = 5 \text{ cm}^2$$

$$a_2 = 2 \text{ mm}^2$$

$$d_1 = 3 \text{ cm}^2$$

$$d_2 = 1 \text{ mm}^2$$

$$d_3 = 2 \text{ mm}^2$$

$$d_3 = 3 \text{ mm}^2$$

$$\epsilon_{r1} = 5$$

$$\epsilon_{r2} = 6$$

$$\epsilon_{r3} = 7$$

Given

$$a_1 = 5 \text{ cm}^2$$

$$= 5 \times 10^{-4} \text{ m}^2$$

$$d = 2 \text{ m}$$

$$\epsilon_0 = 8.854 \times 10^{-12}$$

$$C = \frac{\epsilon_0 A}{d}$$

$$C = \frac{8.854 \times 10^{-12} \times 5 \times 10^{-4}}{2}$$

$$C = 2.2135 \times 10^{-15} \text{ F}$$

$$\frac{d_1}{\epsilon r_1} + \frac{d_2}{\epsilon r_2} + \frac{d_3}{\epsilon r_3} = \frac{1}{5} + \frac{2}{6} + \frac{3}{7}$$

$$= 0.2 + 0.333333 + 0.428571 = 0.761904$$

$$C_1 = \frac{8.854 \times 10^{-12} \times 3 \times 10^{-4}}{0.761904}$$

$$C_1 = 2.761 \times 10^{-15} \text{ F}$$

0/pv
2m